

# Python Basics

Beginner sample document | Learning and document Q&A; practice

## 1. What is Python?

Python is a general-purpose programming language known for readable syntax. It is used for automation, web development, data analysis and artificial intelligence. Indentation groups statements into blocks.

## 2. Variables and data types

A variable refers to a value. Common types include int for whole numbers, float for decimal numbers, str for text and bool for True or False.

```
name = "Anu"
age = 20
score = 87.5
is_student = True
print(name, age)
# Output: Anu 20
```

## 3. Collections

A list is an ordered, mutable collection. A tuple is an ordered, immutable collection. A dictionary stores key-value pairs. A set stores unique elements.

```
courses = ["Python", "Machine Learning"]
student = {"name": "Anu", "course": "Python"}
print(courses[0])
# Output: Python
```

## 4. Conditions and loops

An if statement selects a block based on a condition. A for loop repeats actions for items in an iterable.

```
marks = 75
if marks >= 50:
    print("Pass")
else:
    print("Fail")
for number in range(1, 4):
    print(number)
# Output: Pass, then 1, 2 and 3 on separate lines
```

## 5. Functions

A function groups reusable instructions. Parameters receive inputs, and return sends a result back to the caller.

```
def add(a, b):
    return a + b
print(add(4, 6))
# Output: 10
```

## 6. Files and exceptions

Use with open(...) to manage a file safely. Reading uses mode "r"; writing with "w" creates a file or overwrites its existing contents. try and except handle expected runtime errors.

```
try:
    value = int("hello")
except ValueError:
    print("Enter a valid integer.")
```

## 7. Practice questions

What is Python used for? How does a list differ from a tuple? What does a function return? Write a loop that prints numbers from 1 to 5.